AGENT PRODUCTION BUILD PLAN — Non-LLM Agent Ecosystem (full in-depth report)

> Final deliverable · 2026-10-26 · assembled from ALL planning sessions (S2–S22) + fresh point-by-point web verification (sources in §19). System name: pending owner reveal. This is the production-ready, instruction-based build plan for the complete agent layer: one deep agent design, all ~30 styles, communication, population sizing, memory, training, limits, and the decade-durability architecture.

0 · How to read this report (evidence discipline)

- **Every number is sourced or marked as our own budget/target.** Sourced = §19 evidence appendix with live-verified citations (2026-10-26). "Our budget" = a design target we set, not a measured claim.
- **Honest language:** where the research says something is NOT possible (e.g. precise regime-duration prediction), this report says so instead of selling it.
- **Plain English first, detail underneath.** Tables carry the density.

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1 · Executive summary — what we are building

One line: thousands of tiny, fast, non-LLM specialist agents — one per symbol, one style each — thinking in microseconds inside a deterministic safety cage, trained nightly, coordinated by a message bus, governed by a risk governor that has zero machine learning in it, and all of it wired into seven engines and three rails.

Layer	What it is	Key rule
Cells (the agents)	one symbol × one style, 5–50M-param models, Rust hot path	never touch own limits (R7)
Family trunks (6)	equities · futures · options · crypto · FX · predictions	cross-family transfer forbidden
Engines (E1–E7)	feature · runtime · training · arena · governor · execution · ledger	styles are tenants of engines
Rails (R1–R3)	signal bus (NATS) · memory (LanceDB) · model registry	audited, need-to-know
Safety	deterministic governor, certification gates, crisis replays, human ratification	LLMs never in hot/risk path (R3)

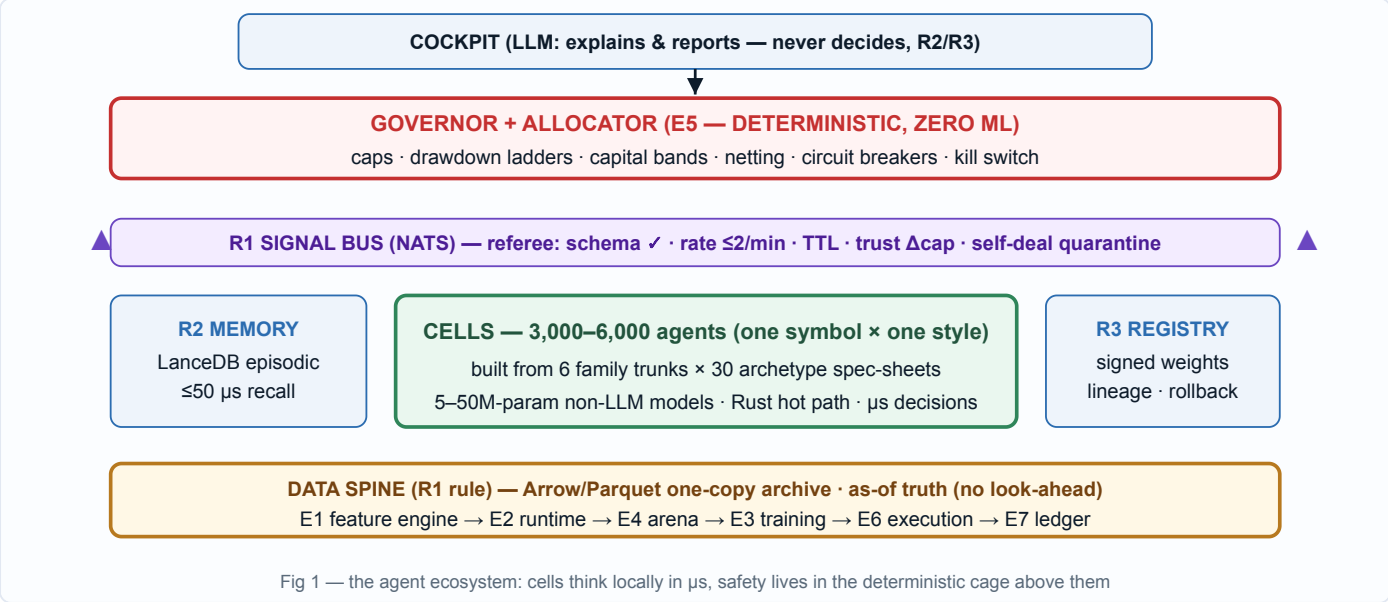


Fig 1 — the agent ecosystem: cells think locally in µs, safety lives in the deterministic cage above them

The four answers people ask first:

1. **Can it find alpha by itself?** Yes — bounded self-discovery inside its spec-sheet IS the agent's job (Sessions 20–21). It can also invent new strategy *combinations* (Invention Lane) — but never new rules.
2. **Is one generalist agent per symbol better?** No — verified against research (negative transfer) and industry (the pod model): specialists win (Session 22).
3. **Will it make profit?** Not promised. Most candidates fail their gates — that is why the gates exist. The plan optimizes for *surviving* reality, not backtest curves.
4. **How long does it last?** Designed for a decade: event-sourced everything, versioned contracts, rolling trunk refresh, swappable edges (§16).

2 · The complete agent — anatomy, deep dive

2.1 The skeleton (identical for every agent of every class)

```
                ┌──────────── OFFLINE (minutes–hours, Python/JAX, never live) ────────────┐
                │  data → features → train (SB3/PyTorch/JAX) → eval-CI → arena race       │
                │  → crisis replays → DSR gate → shadow → HUMAN gate → registry           │
                └──────────────────────────────────┬──────────────────────────────────────┘
                                                   │ signed weights (ONNX) + version
┌──────────────────────────── HOT PATH (Rust, inference-only, µs) ─────────────────────────┐
│ Arrow mmap bus → obs builder (as-of fields) → policy core (trunk+head, ONNX/tract)       │
│ → conformal calibrator → DETERMINISTIC risk gateway (caps/ladders/bans — no ML)          │
│ → Head Contract {action_typed, size, confidence, ttl} → unified order model             │
│ → venue adapter → exchange                                          every hop CI-timed │
└──────────────────────────────────────────────────────────────────────────────────────────┘
        ↑ ring buffer with skip-tags (C4) · every decision logged to the ledger (D3)
```

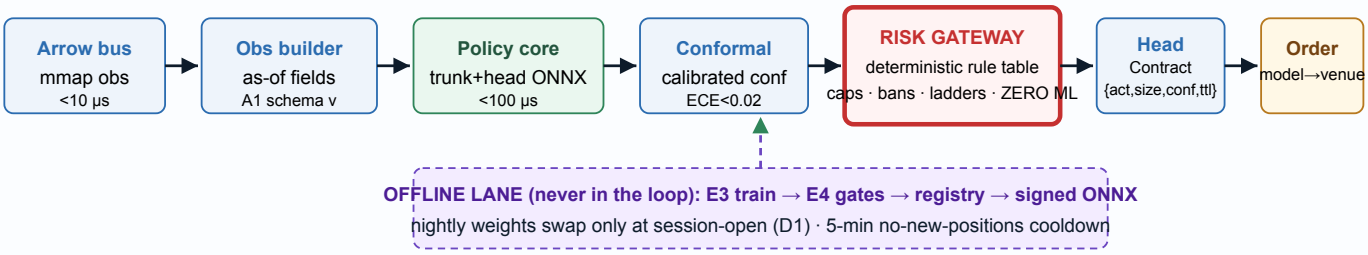


Fig 2 — one decision, start to finish: proposal → calibration → deterministic gate → typed contract. No language anywhere.

What differs between agents is exactly three things: action space, reward formula, auxiliary learned models. Everything else is shared code — **no new codebase per style.**

2.2 Structure — the three-layer identity

Layer	Learns	Lifetime	Example
Family trunk	market physics of ONE family (order-flow shapes, session rhythm, vol structure)	refreshed yearly	the futures trunk
Archetype spec-sheet	the payoff problem (frozen design: obs blocks, action, reward, hazards, champion)	changed only by owner ratification	theta-harvest
Cell instance	that instrument's personality (when IT gaps, fills, fades)	retrained nightly	NIFTY-weekly-theta cell

2.3 How it *thinks* (one decision, start to finish)

1. **Sees** a typed observation (A1): market block (~60 scalars) + position block + signal block (scoped peer signals) + regime block + hazard block. Every field carries an availability timestamp — future data is physically impossible to read.
2. **Measures** its trunk embedding of the current situation and (≤50 µs) recalls the top-8 similar past episodes from its own LanceDB — *measured outcomes only, no narratives* (anti-hallucination by construction).
3. **Decides** with its policy core: a fast neural map obs→action proposal. Output is a *proposal*, never an order.
4. **Calibrates** the confidence: conformal wrapper turns raw scores into honest P(true) intervals (ECE < 0.02 gate per regime).
5. **Gates** through the deterministic gateway: position caps, drawdown ladder, eligibility checks (C1), staleness checks (C4), ban lists, circuit proximity. Zero ML — a fixed rule table.
6. **Emits** the Head Contract and the order model splits it into venue orders (TWAP/ VWAP/POV when size demands).
7. **Records** everything: fills, skips, counterfactual leg P&L → per-cell ledger (D3).

There is **no language, no chat, no free text** anywhere in this loop. That is the whole point (§7).

2.4 The framework stack (what runs where)

Tier	Tool	Runs	Rule
Hot path	Rust + ONNX Runtime / tract	live, µs	inference only, no Python, no GC pauses

Tier	Tool	Runs	Rule
Training	Stable-Baselines3 (reliability harness), PyTorch, pycma (evolution)	nightly GPU	training-time only — deployed artifact is a compiled model file
Research/scale tier	JAX (GPU rollouts, JaxMARL-class)	optional P4	adopt when population > hundreds of cells
Eval	replay arena + DSR/PSR + crisis kits	CI on every merge	seed-invariance, latency CI, adversarial audit
Orchestration	NATS JetStream · PostgreSQL · LanceDB · Arrow/Parquet	always	rails, not brains

2.5 Infrastructure — the seven engines (Session 18)

Engine	Job	Language
E1 Feature	computes obs blocks on the hot path	Rust
E2 Runtime	loads/runs models, conformal calibration	Rust + ONNX
E3 Training	nightly trains/evolves candidates	Python + GPU
E4 Arena	replay evaluation, champion races, crisis kits	Python (deterministic)
E5 Governor	caps, ladders, kill switches, allocator	**Rust, ZERO ML by design**
E6 Execution	TWAP/VWAP/POV + venue adapters	Rust
E7 Ledger	per-cell P&L, reconciliation, tax lane	Rust + Postgres
R1/R2/R3	signal bus / memory store / model registry	NATS / LanceDB / registry

2.6 How it is built (birth → life → death)

```
symbology master picks symbol → data backfill (as-of, PIT-certified)
→ cell BIRTH (A6: ≥20 sessions tape or seeded head from nearest family sibling)
→ reduced capital 0.25× for 30 sessions
→ nightly: E3 trains → E4 races vs champion → crisis replays → DSR gate
→ SHADOW (live decisions, no money) until decision-parity ≥98%
→ HUMAN approval gate → small band capital → monitored → champion races continue
→ underperformer: retirement proposal (B1) → freeze → archive → capital recycled
```

Every step is logged; every gate is a CI job; nothing enters live money without a human signature (R7).

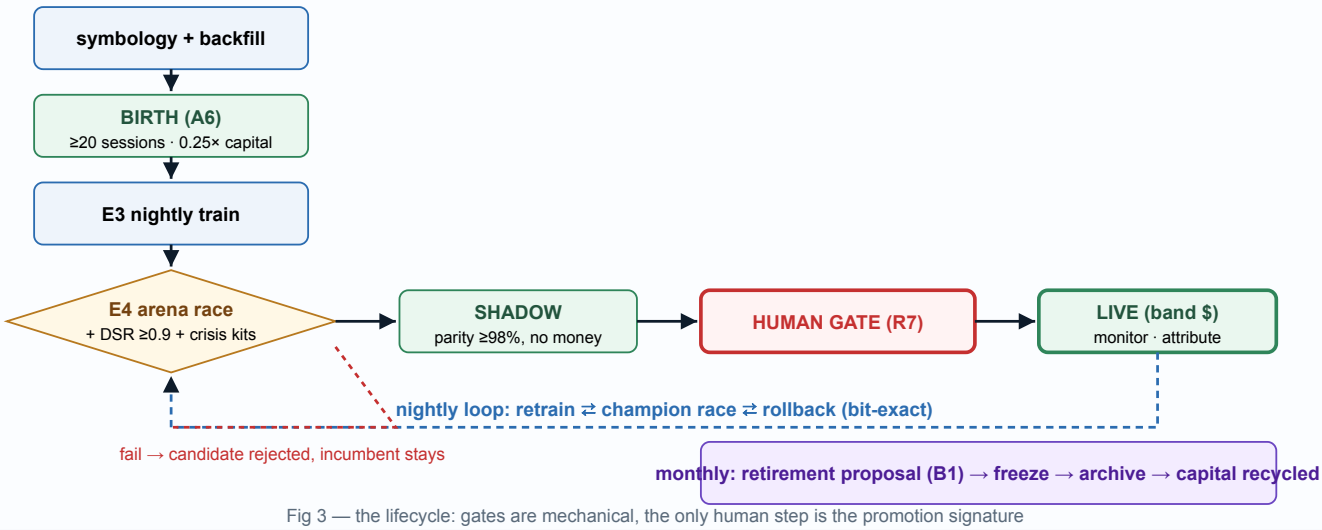


Fig 3 — the lifecycle: gates are mechanical, the only human step is the promotion signature

3 · One symbol, massive data, massive training — how far does it go?

3.1 What “massive data” means per cell (and what the universe actually is)

Verified universe sizes (2026-10-26, §19):

Universe	Count	Source
Listed companies worldwide	**59,581**	WFE Dashboard Oct 2026

Universe	Count	Source
Global equity symbols (incl. ETFs/funds)	~340k+	OneTick coverage (240k equities, 100k ETFs)
Global futures contracts	~190k (140k CME alone)	OneTick / TrendSpider
Listed options (US/global)	**4.3M–6M active**	OneTick / LSEG
Crypto markets	**30,537** across 8,468 coins (3,519 perps)	WorldCoinIndex Oct 2026
Crypto spot/futures/options markets	21k / 10k / 14k	CoinMetrics coverage
FX liquid pairs	~40–70	Stooq major+minor

But we do not need all of them. The liquid-first roster (our actual target): ~2,000–5,000 equities · ~300 futures · top-underlying options chains · ~200 crypto pairs · ~40 FX · ~100–500 prediction markets ⇒ **first-decade population ≈ 3,000–6,000 cells**, full-world coverage is a scaling decision, not a day-one requirement (§13).

3.2 Per-symbol data depth — how deep one agent can pull

One cell can legitimately consume: full tick/LOB history · trades · quotes · funding & open interest (crypto) · earnings/event calendars · sector & index co-moves · cross-venue prints · news-tension features (as a *feature*, not a decision-maker) · corporate-action-adjusted fundamentals · option surface slices (options cells) · liquidity/holiday calendars. All flowing through the **one Arrow spine** — no copies, no re-serialization (DATA-LAYER doc).

3.3 The honest limit: more data ≠ more edge

Three verified truths keep us honest: 1. **Information density decays with horizon** — tick data helps an MFT cell most; daily fundamentals help a carry cell most. Dumping HFT data on a theta cell adds noise, not signal. That is why the spec-sheet picks the obs blocks. 2. **Regime-duration prediction is NOT solved** — a 2026 study (§19) showed survival models trying to predict "how long does this regime last" fail structurally (concordance 0.20–0.39 ≈ random). We therefore predict regime *state* + dwell confidence, never precise duration (§11). 3. **Training scales with compute, not wishes** — FinRL-Podracer trained a NASDAQ-100 agent over 10 years of minute data in **10 minutes on 80×A100** (§19); our nightly window on 1–2 GPUs with 5–50M-param models is minutes-to-hours per cell-batch — plenty, because our cells are small and data is pre-chunked.

3.4 First training (cold start) and nightly retraining

Phase	What happens	Speed evidence
First train	seeded head (A6) + 20 sessions minimum tape → supervised pretrain on replayed outcomes → offline RL fine-tune → arena race	hours on one GPU (small models)
Nightly retrain	sampling ladder: 30d 50% · 90d 30% · 12mo 15% · whole-history 5% (A3) → candidate → gates	minutes–hours; end-to-end retrain orchestration saves 4–6h cycles in production MLOps (§19)
Intraday adaptation	NOT weight changes — config/hazard level changes only (D1: weights = session boundary, always)	sub-ms per-sample online update exists in literature; we deliberately don't use it live
Evolution	pycma mutates feature/param variants nightly; Invention Lane composes strategies (§4)	generational evolution every ~800s shown feasible (FinRL-Podracer)

4 · The self-improvement machine (how it updates, upgrades, versions, and stays sane)

4.1 The four loops (different speeds, different permissions)

Loop	Cadence	What changes	Permission
Fast loop	per event	nothing — inference only	automatic
Learning loop	nightly	weights, feature weights, calibrators	automatic behind gates
Evolution loop	nightly/weekly	parameter variants, feature sets, Invention-Lane compositions	automatic into arena; promotion needs gates + human
Structural loop	monthly/quarterly	spec-sheet proposals, new archetypes, trunk refresh	**owner ratification only**

4.2 Versions — everything is versioned (the answer to "can it has version?")

`obs-schema v` · `feature-registry v` · `reward-spec v` · `weights v` (content-hashed, signed) · `config v` · `lineage manifest` (training vintage, data window, trials count) · `spec-sheet v` . Session-open is the single atomic swap point (D1); any live change forces a 5-minute no-new-positions cooldown; rollback = flip pointer to the previous bit-exact champion. Version numbers ride *inside* the observation so a head can never consume features from a mismatched schema.

4.3 Why it does not hallucinate (the honest engineering answer)

"Better understanding without hallucination" is not a wish — it is a consequence of design choices, each verified against how LLMs fail (§19):

LLM failure	Our countermeasure
free-text reasoning invents facts	**no language in the loop at all** — tensors in, typed contract out
numeric hallucination (a wrong delta/risk figure)	every number comes from deterministic math libraries (Black-Scholes, Kalman, PCA) — the model only *chooses among measured options*
memory contamination (narrative drift)	episodic store holds **measured trade outcomes only** — no summaries, no prose
non-reproducible outputs	content-hashed weights + seed-invariance CI: same seed ⇒ same weights, bit-exact
uncalibrated confidence	conformal calibration gate (ECE < 0.02 per regime) — a number that claims 80% must be right ~80%

4.4 "Consciousness on the data" — the honest reframing

There is no sentience, and we do not need it. What the owner is describing has three real engineering names: **state estimation** (the regime block + hazard block = a continuously updated picture of *what kind of market this is right now*), **attention** (obs-scoping: the spec-sheet decides what matters; the governor strips what the cell may not see), and **calibrated self-knowledge** (the agent's confidence is measured, not felt — and a miscalibrated agent fails its gate). A decade of "understanding" without one gram of consciousness — and every decision remains auditable, which a consciousness could never promise.

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5 · The rules we give the agent (the complete rulebook)

5.1 The eight non-negotiables (R1–R8)

#	Rule	What it forbids
R1	One shared data spine, as-of truth only	look-ahead, duplicate stores
R2	LLMs live ONLY in cockpit/research	any LLM in hot/risk path
R3	Deterministic governor owns risk	learned risk logic, agent-touched limits
R4	Build order follows dependency truth	parallel fantasy work
R5	Certification before capital	unproven candidates trading money
R6	Everything measured, everything logged	silent failure, unauditable decisions
R7	**Agents never touch their own limits**	self-modifying risk, reward hacking of constraints
R8	Human ratification for structural change	autonomous spec-sheet/species evolution

5.2 Per-agent standing constraints (the cage around every cell)

- **Action contract** `{action_typed, size, confidence, ttl}` — nothing else can be emitted; ttl expiry = proposal dies, never executes stale.
- **Capital band** set by allocator; hard caps (max position, max loss/day, max participation, drawdown ladder) enforced by E5 — *before* the order exists.
- **Eligibility gates** (C1): ban lists, circuit proximity, margin class, calendar — governor blocks ineligible instruments with zero discretion.
- **Staleness gates** (C4): drop-tags travel with data; staleness ⇒ cancel quotes, reduce-only; sustained drops ⇒ governor pauses the cell.
- **Hazard gates** per class: expiry rules (C3), spoofing score (C2), poison-tape detector (MM), one-sided pausing.
- **Bus etiquette** (B3): ≤2 signals/min/type, TTL freshness, bounded trust deltas, self-dealing quarantine.
- **Isolation** (B4): a cell NEVER sees other cells' positions/capital/orders — scoped signals only, stripped at the gateway before the cell reads anything.

5.3 What the agent is *allowed* to change by itself

Thing	Self-change?
weights / feature weights / calibrators (nightly, behind gates)	✅ automatic
parameter variants, feature sets (evolution loop)	✅ into arena only
strategy compositions (Invention Lane)	✅ propose → same gates → **human ratifies**
reward formula, risk limits, gateway logic, action contract	❌ never (R7/R8)

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6 · What goes wrong — the complete failure catalog (and the recipe that succeeds)

6.1 The 12 engineered defenses (from the master design)

#	Failure	Defense
1	non-stationarity	regime detectors in obs, rolling windows, DD de-risk ladder
2	overfitting	CPCV + purge/embargo, DSR with trials-logged, parameter budgets
3	look-ahead	as-of-only channel, physically enforced at data layer
4	survivorship	membership-at-date universe, certified else refuses to run
5	data leakage	time+cross-instrument purge, feature-version isolation
6	regime break live	governor capital cut + correct-head swap, dwell confirm
7	reward hacking	independent audit metrics read by committee, versioned reward
8	training instability	seed-invariance CI, grad-clip, NaN-watch, min-stability epochs
9	MM tail inventory	A-S core + bounded nets, auto-widen, one-sided pause
10	execution-cost drift	shadow IS vs sim ±20% ⇒ freeze & recalibrate
11	cross-agent crowding	overlap score + systemic circuit breaker
12	calibration rot	per-regime ECE recheck post-deploy

6.2 "Training itself may go wrong and cause losses" — yes, and here is exactly how

Training failure	How it shows up	What stops it
trained on a lucky window	great replay, awful live	DSR trials-deflation, crisis kits, vintage-matched arena (A5)
learned the wrong regime	confident in the wrong market	regime-stratified eval buckets; preservation gate (A3) — candidate may not degrade older-regime buckets
catastrophic forgetting	forgets 2020 skills after 2026 data	sampling ladder + frozen regime heads + ancient-window CI probe
reward hacking	maximizes metric, destroys money	independent audit metric set (TCA, turnover-adj PnL), committee-read
non-deterministic training	candidate unreproducible	seed-invariance gate — same seed ⇒ same weights or the run is invalid
poisoned data	garbage features	PIT certification, poison-tape detector, backfill QA gates
cost model wrong	sim says profit, live bleeds	MM fills calibrate cost model; ±20% shadow tolerance → freeze

The "successful model" recipe (what a certified agent actually is): expectancy-after-costs positive per regime bucket · DSR ≥ 0.9 with logged trials · beats its deterministic champion on matched folds · survives the 5-crisis replay suite (blind windows) · ECE < 0.02 per regime · shadow decision-parity ≥ 98% · realized cost within ±20% of sim · latency P99 inside budget · human signature. **That list IS the definition of a good agent.** A model trained without it is not a failed model — it is an *untested* one, and untested models never touch money.

6.3 The honest risk statement

Most candidates will fail these gates. Some cells will be retired. No architecture guarantees profit — what this architecture guarantees is that losses stay bounded, explainable, and survivable, and that the system *keeps the winners* (Lo's Adaptive Markets logic: edges decay, the population refills).

7 · Why non-LLM beats LLM for this job (verified, 2026 evidence)

Dimension	LLM agent	Our non-LLM agent	Evidence (§19)
Decision cycle	15–40 seconds	µs–ms internal; MFT decisions at signal cadence	MethodAlgo 2026: 5–6 orders of magnitude gap; <5% of short-term signal value remains after 20s
Cost per decision	~$0.12 inference	~$0 (local compiled model)	MethodAlgo 2026
Numeric honesty	"hallucinates pretty egregiously… extreme errors in quantitative settings"	deterministic math libs; model only selects measured options	DL News 2025 (Allora)
Reproducibility	non-deterministic outputs — MiFID II / SEC 15c3-5 incompatible	bit-exact, seed-invariant, signed weights	MethodAlgo 2026 compliance thesis
Self-revision safety	deployed LLM agent: operator had to remove 11 of 19 self-added rules, median 8-day lag	self-revision limited to weights behind gates; rules are constitution	AllPath 2026 production incident taxonomy
Extreme markets	API rate-limits/outage risk exactly when it matters	local, no external dependency	MethodAlgo 2026
Best role	research proposer, explainability cockpit, attribution narration	**trading principal**	our R2/R3 rule — matches the literature's own recommendation

Where the LLM stays: research-lane strategy proposals (Invention Lane), morning-report narration, owner cockpit Q&A — read-only, never deciding, never in the risk path.

8 · Latency, throughput, accuracy — the numbers (sourced vs ours)

8.1 What the world actually achieves (verified)

System	Tick-to-trade	Notes
NanoARB (Rust, CME HFT)	**median 780 ns · p99 1.2 µs**	LOB 45ns + features 120ns + inference 580ns, ONNX, AMD EPYC (§19)
RustAlgorithmTrading	book updates <5 µs p99 · 100k+ msgs/s	Rust, Ryzen 9
QuantMemo worked example	10 µs full budget (market making)	parse 1.5 + features 3 + model 3.5 + risk 1 + send 1
LLM trading agent	15–40 **seconds**	the contrast that justifies our whole architecture

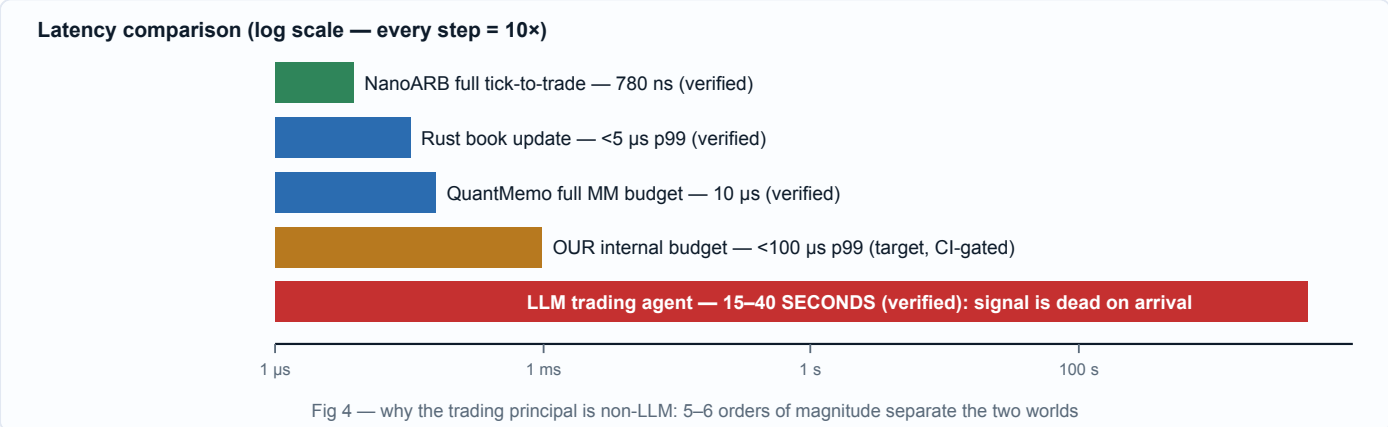


Fig 4 — why the trading principal is non-LLM: 5–6 orders of magnitude separate the two worlds

8.2 Our budgets (targets we set — to be CI-enforced, not yet measured)

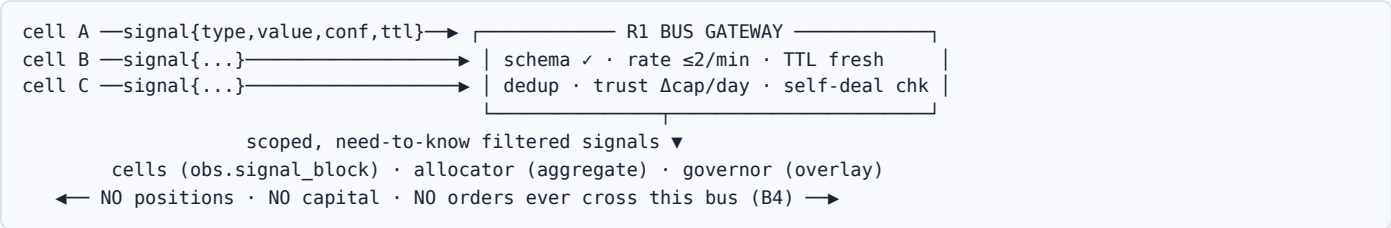
Hop	Our budget (p99)
Arrow mmap obs read	< 10 µs
policy core inference (5–50M params, quantized)	< 100 µs
conformal + deterministic gateway	< 20 µs
order model + venue adapter	< 50 µs
internal total	**< 100 µs** (already written into the master doc's latency-CI gate)
venue hop (network + exchange)	outside our control — measured per venue, priced into cost model

Decisions per second, operations per second (honest math): - Hot loop processes **every book update** — thousands of events/s per liquid cell are trivial at µs each (100k+ msg/s verified in comparable Rust systems). - An MFT cell *acts* orders far less often: signals arrive seconds-to-minutes apart; order rate is bounded by venue rate limits anyway. Speed is for *seeing and gating*, not for spamming orders. - **Accuracy we can promise:** calibration accuracy (a stated 80% must be ~80%) — NOT price-prediction accuracy. Anyone promising prediction accuracy is selling something; our edge metric is expectancy-after-costs per regime (Metrics Bible §5).

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9 · How the agents communicate (and is it better or messy?)

9.1 The design: a refereed signal bus, not an agent free-for-all



What is shared (and why): signed belief signals — {type, value, calibrated confidence, ttl, trust weight} — e.g. “event-cell: earnings surprise incoming, conf 0.71, ttl 40 min”. Enough for a peer to *condition* its own view; never enough to copy another’s book.

What is never shared: positions, capital, orders, limits. Only the governor sees aggregate exposure. The auditor gets a signed, logged exception view.

9.2 Is communication a net positive? (the verdict)

	Without bus (isolated)	With bus (our design)
cross-style coordination (MM hedged by directional view)	impossible	**at the coordinator layer** — visible, auditable
contagion / spam / collusion risk	n/a	bounded by referee: rate caps, TTL, trust deltas, self-dealing quarantine
attribution	clean	stays clean — signals carry source + trust, never merge into one skull
failure mode if a cell goes bad	contained	contained — quarantine freezes its trust, incident logged

Verdict: better, not messy — because the bus carries *claims with receipts*, not conversations. The failure mode of agent-to-agent chatter (emergent nonsense, collusion, feedback loops) is structurally excluded: datagram signals, audited, rate-limited, need-to-know scoped. Evidence for the messaging substrate: NATS handles **millions of messages/second at sub-millisecond latency**, single binary, JetStream adds durable streams + replay + at-least-once (§19) — i.e. the bus is a solved problem at our scale.

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10 · How fast does it learn, adopt, and let go? (adaptation speed)

The agent has **three speeds of change**, each with a verified detector behind it:

Loop	Trigger	Mechanism	Speed (design)
Detect “this is going bad”	continuous	drift detectors on the cell’s own outcome stream: **CUSUM / Page-Hinkley** (fast triggers on small persistent shifts), **ADWIN** (adaptive window compares old vs new averages), DDM/EDDM (error-rate monitors) — the standard, peer-reviewed family, already in our research map (Page 1954)	detector fires within tens–hundreds of observations after a real shift; governor acts on confirmed breach, not flicker
Leave the bad thing	detector + gates	capital ladder steps down (deterministic), hazard level rises, candidate search widens; persistently negative cells → retirement proposal (B1)	same session for capital cuts; days–months for full retirement (with human confirm)
Adopt the good thing	nightly arena	new candidate must beat champion on matched windows → shadow → human gate	best case: ~2 days (nightly train → next-session shadow → approval); never instant, on purpose
Learn a whole new symbol	birth	seeded head + 20 sessions tape + reduced capital 30 sessions	weeks to full band

The honest asymmetry: losing is fast, winning is slow. Cutting risk is a deterministic table lookup (no ML, no approval for *tightening* — D1 monotonic rule: safety may tighten immediately, loosening waits for session boundary). Gaining trust requires passing every gate. That asymmetry is the entire survival strategy.

Evidence note: concept-drift detection is a mature field (CUSUM 1954, ADWIN, PH, KSWIN, BOCD all current in 2026 libraries — §19); the parameter trade-off (false alarm vs missed detection) is real and is tuned per cell class in E4, not hand-waved.

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11 · Regimes: how it knows *what market this is* — and the honest limit on *how long*

11.1 Detection (what we CAN do well)

The regime block in every obs (A1) is fed by a detector stack:

1. **HMM / Markov-switching detectors** (Hamilton 1989) — discrete states (calm / trend / stress / transition) with posterior probabilities, per family.
2. **CUSUM-class fast triggers** — quick flag on mean/variance shifts (the "something changed" alarm).
3. **Structural features** — realized-vol level, correlation breakdown, liquidity depth, cross-asset moves, session phase → a feature vector the detector consumes.
4. **Dwell-time discipline** — the governor only *acts* on a regime state after dwell confirmation (master doc §4.3: allocation changes only when regime state confirms, not on detector flicker).

11.2 Duration — the honest answer to "will this regime run hours, days, months?"

We can estimate persistence, not predict duration. Verified evidence (§19): a 2026 study tried exactly this — Weibull survival models on 54 bearish regimes across 8 asset classes — and failed structurally: concordance 0.20–0.39 (worse than random), flat with more training data, and predictions collapsing to boundary values. Their conclusion: HMM covariates carry **no duration-predictive information** under those models.

What we do instead (and what the industry actually does): - **Stay-probability + hazard shaping**, not point forecasts: "state X has dwell 12 days median historically, current posterior stay = 0.83" → a *confidence-weighted persistence prior*, consumed by the allocator's dwell gate. - **Scenario fan, not a date**: crisis kits + Monte-Carlo-class replays produce a distribution of continuations; capital is sized against the *bad* tail of that fan. - **Cheap insurance**: the drawdown ladder de-risks automatically regardless of what the regime predictor believes. The regime block informs; the ladder decides.

11.3 Dynamic response — "change everything based on the regime instantly"

Instant *trading* response = already in obs (every policy sees regime state every event; hazard gates fire immediately). Instant *structural* response = deliberately NO: strategy swaps, spec-sheet changes, and weight swaps happen at session boundaries (D1). The regime block can flip a cell to reduce-only in milliseconds; it cannot flip the cell's brain mid-session. Regime-chasing mid-session is how systems whipsaw themselves to death — dwell confirmation exists precisely to prevent it.

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12 · The 30 styles — full deep-dive (owner-locked rosters, Session 17)

Reading the table: every row is one archetype = one spec-sheet instantiated on the shared skeleton. Columns: *Payoff physics* (what makes it money) · *Signature obs* (what it watches extra) · *Reward core* (what it optimizes) · *Champion* (what it must beat — deterministic) · *Hazards* (its special gates).

12.1 EQUITIES (5) — trunk physics: auction dynamics, liquidity cycles, earnings calendar

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
intraday-MF	short-horizon momentum/mean-revert on liquid large-caps, many small wins	multi-horizon returns, spread/imbalance, session phase	expectancy after costs (cost coefficients real)	linear/GBM signal + fixed filter	cost drag, regime flicker, news spikes
stat-arb PAIR	cointegrated pair spread reversion, market-neutral	Kalman spread state, hedge ratio, residual stats	spread-PnL (leg ledger A4)	static hedge-ratio OLS	spread break, leg gap, borrow
earnings/event	pre/post-event drift & surprise response	event calendar, surprise history, IV context, news-tension	event-window expectancy	calendar-anomaly baseline	event misspec, gap risk, IV crush
auction (open/close)	open/close imbalance continuation, auction theory	opening auction book, close-auction volume profile	session-window expectancy	time-of-day fixed rule	auction manipulation, one-shot windows
equity MM (where loop permits)	spread capture + queue position	depth×levels, queue, toxicity (VPIN-class)	spread − adverse − inventory − fees	A-S static core	adverse selection, poison tape, quote storms

12.2 FUTURES (4) — trunk physics: term structure, basis, roll, margin cycles

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
delta-direction	trend/conviction on index & commodity futures	multi-horizon momentum, positioning (COT-class), vol regime	directional expectancy after costs	Donchian/MA-class system	whipsaw, gap risk, roll days
basis/calendar-spread	relative value between delivery months	term-structure slopes, spread z-scores, inventory	spread expectancy	static z-score threshold	delivery mechanics, liquidity per leg
term-structure carry	roll-yield capture (contango short / backwardation long)	roll yield, curve slope, seasonality	carry after roll costs	constant-carry rule	curve inversion, regime break
futures MM (tick-aware)	spread capture on liquid contracts	tick book, queue, imbalance	spread − adverse − inventory	A-S static core	tick whipsaw, news halts

12.3 OPTIONS (8) — trunk physics: vol surface, greeks, expiry mechanics (largest roster)

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
delta-direction via options	leveraged directional via premiums	surface deltas, IV vs RV, greeks	direction after premium decay	futures-direction equivalent	IV crush, theta bleed, gap
vol-alpha (vega)	trading implied vs realized vol, term-structure of vol	IV surfaces, RV forecasts, skew	vol spread expectancy	straddle-straddle static rule	vol regime jumps, gap IV
theta-harvest	premium decay capture (short premium, defined risk)	term IV rich/poor, event proximity	decay after tail costs	short-strangle baseline	fat tails, gap events (D2 kits mandatory)
gamma-scalper	long gamma day-scalp vs decay	gamma/theta ratio, intraday RV	scalp PnL − decay	no-gamma control	decay drag in calm markets
basis/calendar (options)	calendar/diagonal spread value	term-IV shape, forward vol	spread expectancy	static calendar rule	pin, early assignment
expiry/pin specialist	pin-risk capture near expiry	OI walls, dealer gamma, pin score	expiry-window expectancy	no-expiry baseline	pin breaks, settlement rules (C3)
hedge-executor	cheapest hedge for the book (cost minimization)	greek exposures, hedge ratios, borrow	hedge cost vs slippage (NOT alpha)	static rebalance rule	hedge slippage, leg selection
options-MM (vol-aware quoting)	two-sided vol quoting, inventory control	surface microstructure, greeks P&L	spread − adverse − vol-inventory	A-S on surface	surface jumps, adverse selection

12.4 CRYPTO (6) — trunk physics: 24/7 sessions, funding, liquidations, fragmented venues

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
perp funding-rate	funding capture on perp basis	funding rate, basis, OI	funding + basis after fees	static funding rule	funding flips, venue risk
liquidation-cascade hunter	fade/ride forced-flow cascades	liquidation heatmap, OI drops, leverage clusters	cascade-window expectancy	level-breakout baseline	fakeouts, exchange outages
weekend-session quirks	weekend liquidity/behavior patterns	session-phase (crypto), weekend flags	weekend expectancy	weekday-only baseline	thin books, gap risk
stat-arb PAIR (crypto)	cross-pair cointegration	pair spreads, venue basis	spread expectancy	OLS hedge	de-peg events, 24/7 drift
cross-venue MM	venue price dispersion, transfer flow	multi-venue books, transfer alerts	spread − adverse − transfer cost	single-venue A-S	venue outage, deposit delays
perp MM	perp spread capture	perp book, funding clock, OI	spread − adverse − inventory	A-S static core	funding-time adverse, halts

12.5 FX (3) — trunk physics: session overlaps, macro calendars, carry

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
session-pattern	London/NY/Asia session behaviors	session phase, overlap liquidity, fixings	session expectancy	time-of-day rule	session regime shifts
carry	interest-rate differential	rate differentials, vol regime, risk-on/off	carry after vol drag	static carry basket	carry unwind (crisis kits)

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
news-fade	overreaction fade after macro prints	calendar, surprise index, post-news path	fade expectancy	post-news drift baseline	twin-fire slippage, gap

12.6 PREDICTIONS (4) — trunk physics: resolution risk, 0/1 payoffs, no continuous book (own trunk, day one)

Style	Payoff physics	Signature obs	Reward core	Champion to beat	Key hazards
probability-alpha	mispriced event probabilities vs model	market prices, base rates, resolution data	Brier-improvement → PnL	base-rate estimator	resolution risk, thin books
event-odds arb	same event, price discrepancy across markets	cross-market quotes, correlation of events	arbitrage spread after fees	naive cross-quote rule	one-leg unfillable, resolution mismatch
resolution-discount	trading resolution certainty drift	resolution timeline, rule text, volume	expectancy on certainty convergence	calendar baseline	rule ambiguity, delisting
probability-MM	two-sided quoting on event odds	book depth, resolution clock	spread − adverse − resolution risk	A-S adapted to 0/1 payoff	resolution jumps, adverse selection

Cross-cutting standing elements: Type-3 portfolio layer (allocator + governor over ALL families), execution module (TWAP/VWAP/POV), event triggers (cross-family OBSERVATION triggers, not a fifth family).

Adoption waves (anti-explosion rule): W1 equities core → W2 event + first derivative trio → W3 MM + options depth → W4 crypto rosters → W5 FX + predictions. An archetype ships ONLY when its family trunk exists ∧ deterministic control is implementable ∧ ops can absorb its shadow slot.

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13 · How many agents do we actually need? (population sizing, with math)

13.1 The world universe vs our roster (honest funnel)

Tier	Definition	Count (evidence §19)
Full world universe	every listed equity (WFE) + futures + options + crypto markets + FX + prediction markets	59.5k companies · ~190k futures contracts · 4.3–6M options · 30.5k crypto markets · ~60 FX
Tradable-for-us universe	liquid enough for our cost model, in markets we choose to connect	see below
Our first-decade roster	liquid-first, per family, gated by venue decisions	**≈ 3,000–6,000 cells**
Day-one slice (W1)	1–2 symbols × equities styles + pilot portfolio layer	2–10 cells

Liquid-first roster math (our estimate, clearly labeled):

Family	Selection rule	Cells
Equities	top ~2,000–5,000 by ADV in chosen markets	2,000–5,000
Futures	top ~200–300 contracts by open interest	200–300
Options	top ~30–50 underlyings × chains treated as one cell each	30–50
Crypto	top ~150–250 pairs/perps	150–250
FX	liquid majors+minors	~40
Predictions	active liquid markets on chosen venues	100–500
Total		**≈ 3,000–6,000 cells**

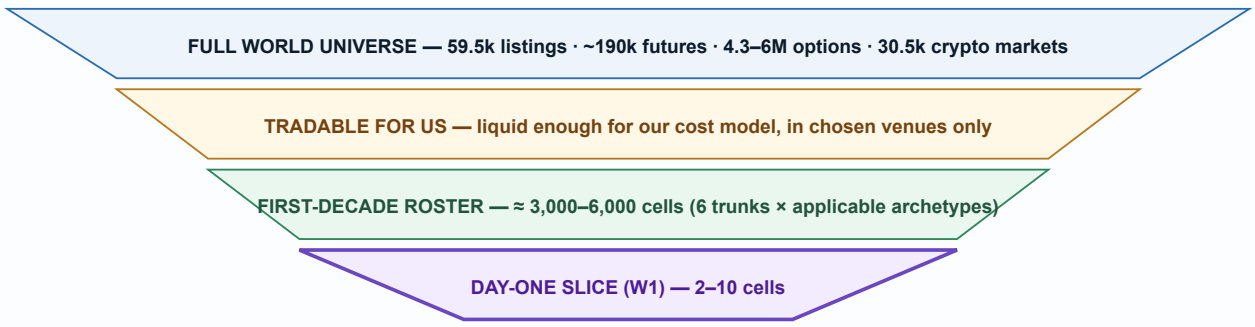


Fig 5 — the population funnel: we start tiny, grow through wave gates, never "deploy the world" at once

Key insight — cells ≠ styles × symbols fully crossed. Not every style applies to every symbol (the roster matrix in §12 defines which cells exist). A cell is born only where liquidity, data, and a deterministic champion exist.

13.2 What one cell costs (scale economics, P4 numbers to confirm)

- Model: 5–50M params → ~20–200 MB with versions in registry (est.).
- Episodic memory: ≤100k episodes × ~1 KB → ≤100 MB/cell (A2 ceiling).
- Training: batched — dozens of cells per GPU-night; small models, minutes each.
- Hot path: µs inference, near-zero marginal cost. ⇒ **Thousands of cells are operationally affordable on a small GPU fleet + 1–2 boxes** — this is the whole reason for tiny shared-skeleton models instead of giant ones.

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14 · Orchestration — how the groups are run over markets

14.1 The orchestration stack (exactly four levels, not one per market)

```
L4  OWNER / COCKPIT          — human gates, morning report, kill switch (LLM explains, never decides)
L3  GOVERNOR + ALLOCATOR     — ONE per system (E5): caps, ladders, bands, netting, circuit breakers
L2  FAMILY TRUNKS (6)        — trunk refresh, shared feature/encoder training, champion per class
L1  CELL OPERATORS           — per-market runtime groups: feed adapters, session managers,
                               batch schedulers (one operator covers a market's cells)
L0  CELLS (3,000–6,000)      — the agents themselves
```

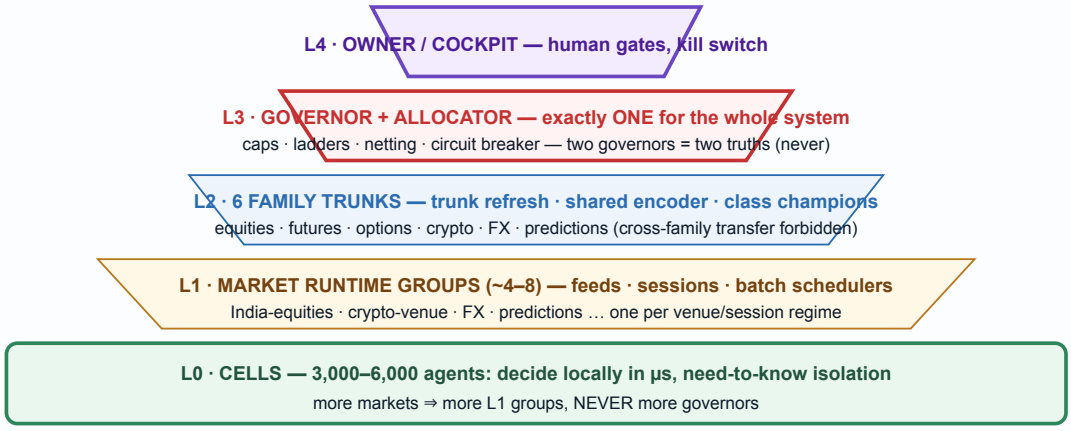


Fig 6 — orchestration: four levels, one governor, deterministic scheduling (no "herding minds" — there is no language)

How many orchestrations? Not one per market — **one governor (L3) for the whole system**, six trunk managers (L2), and a handful of *market runtime groups* (L1: e.g. India-equities runtime, crypto-venue runtime, FX runtime, predictions runtime — one per venue/session regime, ~4–8 groups initially). More markets = more L1 groups, NOT more governors. The governor must stay singular — two governors = two truths about risk (that is exactly how pod shops keep a *central* risk desk).

Does orchestration make the agents better? The cells don't need orchestration to think — they decide locally in µs. Orchestration adds what local thinking cannot: capital allocation across cells, netting of conflicting exposures, contagion control, signal routing, and fleet-wide training scheduling. Since our agents are non-LLM tensors (not chatty personas), orchestration is *scheduling + arbitration* — deterministic, cheap, and auditable — not "herding minds". That is why it stays clean at thousands of cells: **messiness in agent orchestration comes from language; we have none.**

14.2 Fleet operations rhythm

Cadence	Job
per event	cells decide; governor gates; ledger records
per session open	atomic swap point (weights+features+config), eligibility recompute
nightly	E3 trains batch → E4 races → registry updates → morning report
weekly	evolution loop + Invention Lane sweep; diversity/monoculture check (B2)
monthly	retirement proposals (B1), covariance refresh, crisis-suite audit
yearly	trunk refresh, spec-sheet reviews, suite expansion (D2), tech-refresh review (§16)

15 · The memory layer for thousands of agents (long-term → short-term, organized)

15.1 The four memory tiers (per cell + shared)

Tier	What it holds	Where	Retention	Recall speed
T0 working state	current obs, recurrent slot, position state	in-process RAM (mmap)	session-lived	~µs (same process)
T1 episodic	measured trade situations → outcomes, regime-tagged	per-cell embedded LanceDB	age-decayed + regime-stratified, ≤100k episodes/cell	top-8 ≤50 µs budget (A2)
T2 training corpus	full tick/bar history, as-of, PIT-certified	Arrow/Parquet archive (L0, immutable)	**forever** (hash-chained)	columnar scan, offline only
T3 fleet knowledge	weights, lineage, champions, spec-sheets, feature registry	model registry + Postgres	forever, versioned	ms

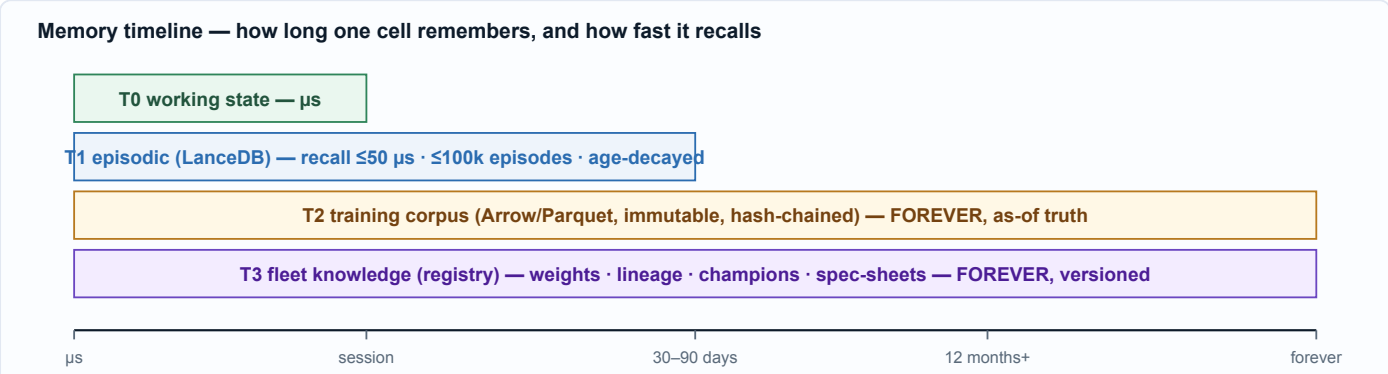


Fig 7 — four tiers: recent life is cheap and fast, history is immutable, knowledge is versioned · writes are write-behind (hot loop never waits)

15.2 How memories are organized and how long they last

- **Short-term (seconds–days):** T0 + the sampling ladder's newest-30-days bucket — the cell is *most tuned* to the last month (50% of nightly training weight).
- **Medium-term (weeks–months):** 90-day (30%) and 12-month (15%) buckets + episodic top-8 recall of similar past situations.
- **Long-term (years–forever):** whole-history 5% reserve, regime-stratified archive (T2), ancient-window CI probe (a frozen set of old windows re-run on every candidate so the cell can never silently forget how 2020 worked), and the archive itself.
- **Write path:** async write-behind — the hot loop NEVER waits on memory writes; every skip is tagged, every outcome lands in the ledger then the episodic store.
- **Why no hallucination in memory:** entries are structured measurement records {situation-embedding, outcome, regime, event-tag} — there is nothing to "imagine"; retrieval is nearest-neighbor over numbers.
- **Fleet math:** 5,000 cells × 100 MB episodic ≈ 500 GB (one disk) · T2 archive scales with tick volume (TB-class, commodity storage) · T3 registry GB-class. Nothing here needs exotic infrastructure.

16 · The decade-durability architecture (run a decade without rewriting)

Honest premise: no architecture survives a decade unchanged — *what* survives is a stable core with swappable edges. The design below is the pattern that has actually lasted in production systems (event sourcing + versioned contracts + rolling refresh), verified against schema-evolution research (§19: event-sourced systems evolve via versioned events; the known challenges are evolution discipline and projection rebuilds — both engineered for here).

16.1 What NEVER changes (the decade-stable core)

Stable element	Why it lasts
Arrow/Parquet data spine + as-of truth (R1)	format is an open standard; data outlives any code
Event-sourced everything (decisions, fills, config changes as append-only events)	history is reconstructable forever; new views = new projections, no migration of the past
Head Contract shape `{action_typed, size, conf, ttl}`	typed payloads absorb new action kinds without breaking consumers
Deterministic governor concept (R3/R7)	it is a rule table — rules are edited by humans, versioned, never learned
Spec-sheet identity (family × archetype × cell)	new styles = new spec-sheets, zero new codebases
Certification gates (DSR, crisis kits, shadow, human gate)	they are process, not code fashion

16.2 What rolls (the refresh lanes)

Lane	Cadence	Mechanism
Weights	nightly	registry pointer flip, bit-exact rollback
Trunk encoder	yearly (or drift-triggered)	full-window retrain; heads adapt via ladder
Obs/feature schema	quarterly-ish	versioned contracts + compatibility check at session swap (D1)
Venue adapters	when venues change	adapter interface frozen; implementations replaced
Training framework tier	as needed (SB3 → JAX tier at scale)	training-time only — **deployed artifact format (ONNX) never changes**
Crisis suite	yearly expansion	new kits added, old never removed
Hardware/infra	as needed	hot path is plain Rust — no vendor lock in the loop

16.3 The anti-rot mechanisms

- **Doc-drift checks** (standing policy): planning docs → engineering docs with review cadence; the remainder map (P1–P4) is the living gap list.
- **Deprecation discipline:** every versioned contract carries a migration window; the atomic session swap means at most ONE version skew exists at any moment.
- **The ONNX firewall:** Python/framework churn can never touch the live loop — only compiled model files cross the boundary. Frameworks rise and fall; the artifact format is the wall.
- **Blind crisis replay + seed-invariance + latency CI run on every merge** — a decade of merges cannot silently degrade the system because the gates are mechanical.

Claim we will make (after M1 measurements, not before): internal p99 < 100 µs, reproducible training, 24/7 operation with session-boundary swaps. Until measured, these are budgets — the latency-CI gate exists precisely so the claim becomes a fact or gets corrected.

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17 · Advanced technology to adopt (the "even better" menu — each evidence-checked)

Technology	What it gives us	Status
Small sequence models for LOB (Mamba/SSM-class)	NanoARB runs a ~500K-param Mamba LOB model at <800 ns — proof that tiny sequence trunks are production-viable	candidate trunk architecture for E4 shootout (GBM vs temporal-conv vs transformer vs SSM)
GPU-accelerated rollouts (JAX-LOB / JaxMARL-class)	up to **240× faster training** with 4096 parallel envs on one GPU — makes crisis-replay + evolution sweeps cheap at fleet scale	P4 adoption when population > hundreds of cells
Generational evolution ensembles (FinRL-Podracer)	populations of agents evolved in generations, 3–7× faster than RLlib, proven on NASDAQ-100	maps directly to our pycma evolution loop
Inference-time policy conditioning (FinPILOT-class, arXiv 2605.12653)	a forecaster's trajectory adjusts the policy AT DECISION TIME without retraining — compatible with any pre-trained agent	research-lane experiment (P4), never auto-deployed
World-model trial	learned environment model for planning/eval rollouts	already on the P4 research agenda; gated like any candidate
Meta-labeling (LdP)	second model sizes positions on primary-signal confidence — turns our conformal confidence into sizing directly	fold into reward/feature spec (S15)
Foundation forecasting models (TimesFM-class)	generic forecaster as a *feature candidate* to test, never a decision-maker	research lane only (Session 20 verdict)

Technology	What it gives us	Status
Digital-twin LOB simulator (JAX-LOB-class)	high-fidelity replay for E4 + sim-fidelity calibration loop (P3-13)	build in Stage sim-fidelity lane
Conformal prediction (already core)	distribution-free calibrated confidence — the Head Contract's number	adopted, non-negotiable
Drift-detector family (CUSUM/PH/ADWIN — already core)	the "learn fast, leave fast" machinery of §10	adopted, per-cell tuning in E4

Adoption rule: every item on this menu enters through the same gates as any candidate — no technology deploys because it is new; it departs when it beats the incumbent on matched windows.

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18 · Build instructions — the concrete order (dependency-true, with done-when gates)

> Maps to FULL-SYSTEM-PLAN's 10 stages and the P1–P4 remainder map. Each step has a **done-when** gate; nothing starts before its predecessor's gate passes.

Phase A — foundation (unblocks everything)

#	Build this	Done when
A1	**Stage-1 foundation spec freeze** (P1-1): event contract final fields, source-registry schema, storage-edge schemas	spec reviewed + committed; every later doc references it
A2	**Data spine**: Arrow/Parquet archive + as-of() API + hash-chaining	backfill of first symbol passes PIT certification tests
A3	**Connector framework** (P1-2): typed template, checkpoint table, QA gates	one live connector produces certified data end-to-end
A4	**Feature spec + Reward spec** (P1-3,4): actual feature list per class, frozen reward formulas with real cost coefficients	both docs frozen; obs blocks implementable from them

Phase B — the agent machine (one cell, end to end)

#	Build this	Done when
B1	**E1 feature engine** (Rust, hot path) + **obs schema A1** with as-of fields	feature outputs match offline reference bit-exact in CI
B2	**E2 runtime** (ONNX load, conformal calibrator, gateway scaffold) + **latency CI** (p99 budget per merge)	latency CI green on target hardware
B3	**E3 training** (SB3 harness, seed-invariance CI, sampling ladder)	same seed ⇒ same weights, twice, in CI
B4	**E4 arena**: champion race, CPCV+DSR gates, crisis-kit replays (start with 2 kits)	a synthetic candidate can be promoted/rejected by gates alone
B5	**E5 governor**: deterministic rule table, drawdown ladder, kill switch (zero ML)	replayed crash suite: ladder fires correctly, 100% deterministic
B6	**First cell (M1)**: 1 symbol × 1 style — replay train → shadow → human gate 1	shadow decision-parity ≥98%; owner signs gate

Phase C — the loop closes (money-path + coordination)

#	Build this	Done when
C1	**First market decision** (P1-5): India vs crypto vs FX → venue adapter + cost model + tax lane	live tiny-order in chosen venue, full ledger attribution
C2	**E6 execution** (TWAP/VWAP/POV) + **E7 ledger** + morning report	IS within model ±20% on live shadow
C3	**Rails**: R1 signal bus with referee (B3/B4), R3 model registry, R2 LanceDB episodic	bus abuse tests pass (rate, TTL, self-deal quarantine)
C4	**M2 portfolio lite**: static allocator + governor overlay + risk targeting	de-risk ladder correct in replayed crashes

#	Build this	Done when
C5	**Population growth W1→W5** per taxonomy waves	each new archetype passes wave rule (trunk ∧ control ∧ ops)

Phase D — hardening & scale (P3/P4)

Sim-fidelity calibration loop · eval-CI harness golden datasets · full crisis suite (5 kits) · adversarial audit (C2) · capital-allocation scheme + governor table v2 · monitoring/SLO/pager runbook · incident runbooks · explainability cockpit (LLM lane) · scale economics review → JAX tier decision.

Day-one practical commands (scaffolding, when Stage 1 starts):

```
cargo new --lib hot-path          # E1/E2/E5/E6/E7 in one workspace, Rust 2024
uv init --python 3.12 train-lane  # E3/E4 Python lane (uv + ruff + pytest)
# gates: cargo clippy -D warnings · cargo test · uvx ruff check . · uv run pytest
# CI per merge: latency budget test · seed-invariance · DSR gate · secrets scan
```

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19 · Evidence appendix (every external claim → source, all live-verified 2026-10-26)

Universe sizing 1. WFE Dashboard Oct 2026 — 59,581 listed companies (domestic+foreign): focus.world-exchanges.org/issue/october-2026/dashboard 2. OneTick coverage — 240k equities, 100k ETFs, 190k futures, 4.3M options: onetick.com/market-data-coverage 3. TrendSpider — 61k assets, 28k crypto pairs, 140k CME futures contracts: help.trendspider.com/kb/data-feeds/market-data-sources 4. WorldCoinIndex (2026-10-26) — 30,537 markets, 8,468 coins, 3,519 perpetuals: worldcoinindex.com/exchange 5. CoinMetrics — 21k+ spot, 10k+ futures, 14k+ options markets: gitbook-docs.coinmetrics.io/market-data/market-data-overview 6. LSEG — 6M+ active options, 160+ futures feeds: lseg.com/en/data-catalogue/derivatives

Latency / throughput 7. NanoARB — tick-to-trade median 780 ns, p99 1.2 µs (LOB 45ns, features 120ns, inference 580ns, ONNX, AMD EPYC): mintlify docs dhir1007/nanoARB 8. RustAlgorithmTrading — 100k+ msgs/s, book updates <5 µs p99: github.com/SamoraDC/RustAlgorithmTrading 9. Quant Memo — 10 µs tick-to-trade budget worked example + P99 methodology: quantmemo.com/concepts/inference-latency-budgets 10. Inference-latency-budget skill — P99 measurement discipline (warm-up, GC, quantization caveats): skills.himanshujangir.com

LLM limits (the non-LLM case) 11. MethodAlgo 2026 — LLM decision cycle 15–40 s vs <100 µs HFT; ~$0.12/decision; <5% signal value at 20 s; MiFID II/SEC reproducibility conflict: methodalgo.com/press/research/latency-of-logic 12. DL News 2025-10-05 — “LLMs hallucinate pretty egregiously… extreme errors” in quant settings (Allora): dlnews.com/articles/defi/ai-agents-are-terrible-at-trading-crypto-but-that-could-change/ 13. AllPath (arXiv 2610.26662, 2026) — deployed LLM trading agent: operator removed 11 of 19 self-added rules (median 8-day lag); deterministic code gates required 14. Agentic Trading survey (arXiv 2605.19337, 2026) — memory hallucination = critical vulnerability; LLM should be read-only 15. Execution-realism scoping review (arXiv 2606.08285, 2026) — LLM trading evaluation practice immature; costs/survivorship/look-ahead routinely omitted 16. Qlumina 2026 — probabilistic text vs deterministic capital: non-deterministic outputs incompatible with reproducible execution

Training speed 17. FinRL-Podracer (arXiv 2111.05188) — agent trained in 10 min on 80×A100 (NASDAQ-100, 10y minute data); 3–7× vs RLlib; generational evolution every 800 s 18. JaxMARL-HFT (arXiv 2511.02136) — GPU rollouts, 4096 parallel envs, up to 240× training speedup 19. Inferensys MLOps — sub-ms per-sample online updates / 1–5 s retrain cycles; EOD orchestration cuts 4–6 h retraining windows: inferensys.com 20. dTrader — nightly fine-tune on 30 days of trades + 60 s PPO rollout horizon: theclevermachine.com

Regime / drift 21. Regime Survival Forecasting (TQFB 2026) — Weibull AFT fails: concordance 0.20–0.39, flat vs data volume, boundary-collapsed predictions: tqfb.reapress.com/journal/article/view/85 22. CUSUM/Page-Hinkley/ADWIN/DDM/KSWIN/BOCD — current standard drift-detector family: CRAN datadriftR (2026), frouros/river docs; Page 1954; Biometrika 2024 CUSUM 70-year review 23. Hamilton 1989 (Markov-switching) · Lo 2004 (Adaptive Markets) · Bailey–LdP (PSR/DSR/PBO/CPCV) · conformal (Vovk/Angelopoulos-Bates) — the classics already mapped in AGENT-ECOSYSTEM-MASTER §6

Bus / memory / durability 24. NATS — millions msgs/s, sub-millisecond latency, single binary; JetStream streams/consumers/replay/at-least-once: docs.nats.io, nats.io/about 25. ACM TOIS 2025 — memory-mechanism survey for agents (structured review): dl.acm.org/doi/full/10.1145/3748302 26. Event-sourced systems schema evolution (arXiv 2104.01146) — industry evidence: evolution + projection rebuild are THE challenges → our versioned-contract discipline

Specialist-vs-generalist (Session 22) 27. ScaleZero (ICLR 2026) — multi-task negative interference; generalist ≈ or < 26 single-task specialists 28. Pod-model industry structure — Millennium/Citadel/Point72 central-risk-desk architecture (TD/CAIS/Long Angle 2025–26) 29. Quant taxonomy — quant = method umbrella incl. market making, stat-arb, vol (Quantt 2026, The Wick 2026, Springer 2025)

Research-lane candidates 30. FinPILOT (arXiv 2605.12653) — inference-time MPC-style policy conditioning on forecasts

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20 · Open forks (owner decisions still pending)

1. **Demo milestone** — what counts as the first demo.
2. **First market** — India vs crypto vs FX (decides venue adapter, cost model, tax lane).
3. **Engine shootout** — GBM vs temporal-conv vs small transformer vs SSM trunk.
4. **Final system name** — pending owner reveal.
5. **Timeframe sub-type field** — proposed config field, owner verdict pending.
6. **Colab-as-E3-lane** — T4 verified here; decision pending (training lane only).

7. **Long-horizon allocator cell** — P4 capital-allocation session.
8. **Master-doc sync** — AGENT-ECOSYSTEM-MASTER needs family×archetype update + re-render.

End of report. Companion docs: FULL-SYSTEM-PLAN · AGENT-ECOSYSTEM-MASTER · DATA-LAYER-ULTRA-LOW-LATENCY · ROW3-SELF-LEARNING-AGENT-STACK · AGENT-RL-FRAMEWORKS-LIFECYCLE · AGENT-CLASS-DESIGN-TYPE1-2-3 · AGENT-TAXONOMY-FAMILIES · AGENT-17-TOPICS-DESIGN · PLANNING-CONVERSATION-LOG (S2–S23).